

$$a_1 = \cancel{f(z_1)} = \frac{e^{z_1}}{\sum e^{z_i}} \quad 0 + 1 \quad \underline{\underline{0 + \infty}}$$

$$a_2 = \cancel{f(z_2)} = \frac{e^{z_2}}{\sum e^{z_i}} \quad 0 + 1$$

$$a_3 = \cancel{f(z_3)} \quad \vdots$$

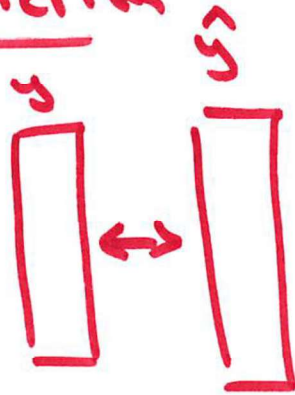
$$a_4 = \cancel{f(z_4)} \quad \vdots$$

$$a_5 = \cancel{f(z_5)} \quad \vdots$$

$$\checkmark \frac{\cancel{\sum e^{z_i}}}{\cancel{\sum e^{z_i}}} = 1$$

$$e^{z_1} \quad e^{z_2} \quad \dots \quad e^{z_5}$$

Loss Function



Loss Func $\rightarrow$ distance measure

$\downarrow$
a 'norm'

Cross entropy

binary classification

$$\frac{1}{n} \sum |y_i - \hat{y}_i|$$

$\rightarrow$ MAE

SAE

$\rightarrow$ MAD



$$\frac{1}{n} \sum (y_i - \hat{y}_i)^2$$

$\downarrow$

SSE

MSE

RMSE

Key

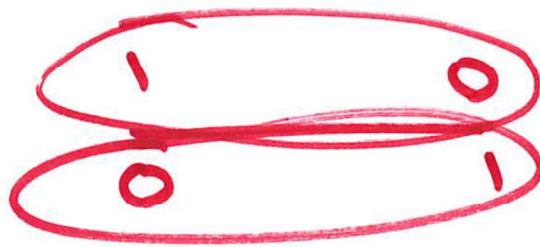
Cross
entropy

$$L = - (y_i \ln(\hat{y}_i) + (1-y_i) \ln(1-\hat{y}_i))$$

$$L = \begin{cases} -\ln \hat{y}_i & \text{if } y_i = 1 \\ -\ln(1-\hat{y}_i) & \text{if } y_i = 0 \end{cases}$$

truth

pred



$$(\hat{y} - y)^2$$

→ Reg → not a big deal
cause you are
1 off. (1-0)²

bin

→ Clam → $-\ln(0) \rightarrow \infty$

$-\ln(1-1) \rightarrow \infty$

1
①

0.9
⑥.4

~~($\hat{y} - y$)²~~